

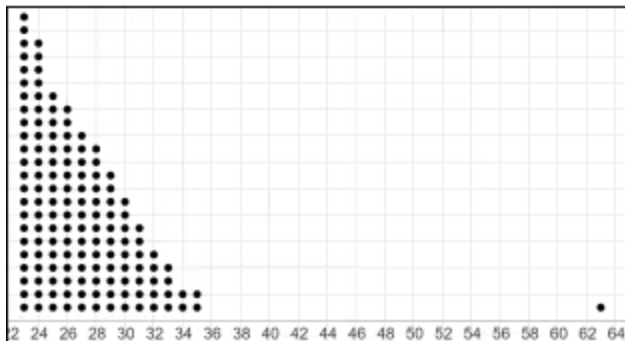


Evaluating the range

1 Which of these sentences best describe what an outlier is. Tick **2** correct answers

- ☐ An outlier is always the largest value in a data set.
- ☒ An outlier is extremely large or small compared to the rest of the data set.
- ☐ An outlier is an extremely large or small value after a range is calculated.
- ☐ Outliers lie close to all the other data points in a data set.
- ☒ Outliers lie far away from where the majority of the data points are clustered.

2 What is the value of the outlier on this dot plot? Tick **1** correct answer



- ☐ 23
- ☐ 35
- ☐ 62
- ☐ 62.5
- ☒ 63

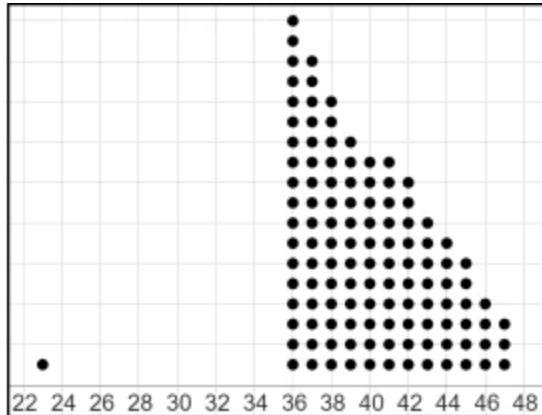
3 How many outliers are there on this dot plot? Fill in the blank



4



4 Match each piece of information to its value. Write the correct letter in each box



a	The outlier	<i>b</i>	24
b	The range with the outlier	<i>a</i>	23
c	The range without the outlier	<i>c</i>	11
d	The largest data point	<i>d</i>	47

5 For this ordered data set, calculate the range whilst including all outliers, and also calculate the range without the outliers. The difference between the two ranges is 83 Fill in the blank

4	5	34	34	34	34	35	35	35
36	36	36	37	37	37	37	37	37
38	39	39	39	39	40	40	40	40
41	41	41	41	41	41	41	43	96



- 6 Different outliers may need to be: kept, removed, investigated further, depending on the context with which these outliers have occurred. Match what should happen to the outlier to its context. Write the correct letter in each box

a	Definitely keep
b	Definitely remove
c	Investigate further

<i>b</i>	A student walked 200 km to the supermarket.
<i>c</i>	19 cats had a mass of 4 - 6 kg. Outlier had a mass of 11 kg.
<i>a</i>	An outlier of 28 for number of days in a month without rain.

